K-FIELD BULLET TRAJECTORY MODULATION

Maximum spin-coupled lateral force and time-of-flight integration for a 90-grain 5.56 mm projectile

Model configuration: 1:6 barrel twist, 2910 ft/s muzzle velocity, and optimal locked Earth-CMB K-Field orientation

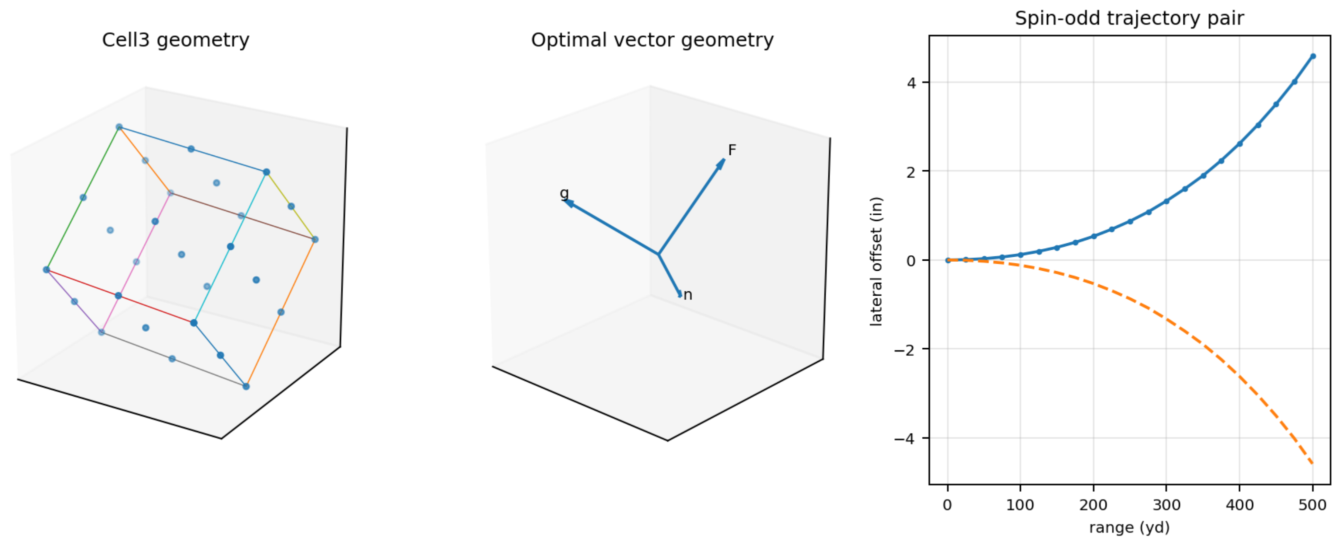

Figure 1. Title composite generated from the authoritative Cell3 geometry, locked field-response vectors, and the supplied ballistic time-of-flight data.

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Archival foundational education reference - Unix epoch 1785151191 - 2026-07-27T11:19:51Z

Frontispiece: authoritative base cell, planes, and all nodes

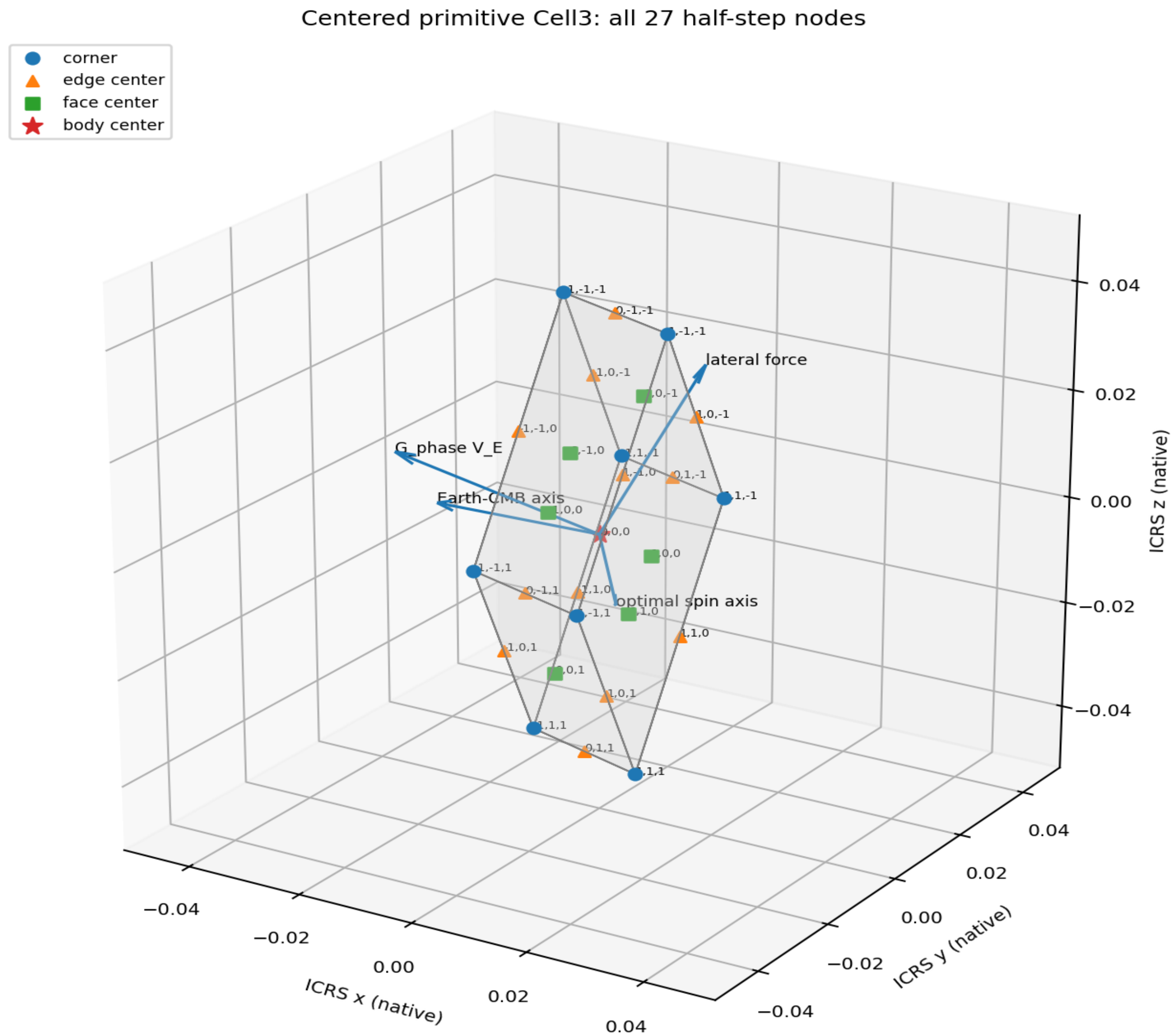

Figure 2. Dimensionally correct centered primitive Cell3 in ICRS native coordinates. The six boundary faces, twelve edges, twenty-seven half-step nodes, integer addresses, Earth-CMB direction, transformed field response, optimal spin axis, and lateral-force direction are shown from source arrays.

Principal Cell3 planes

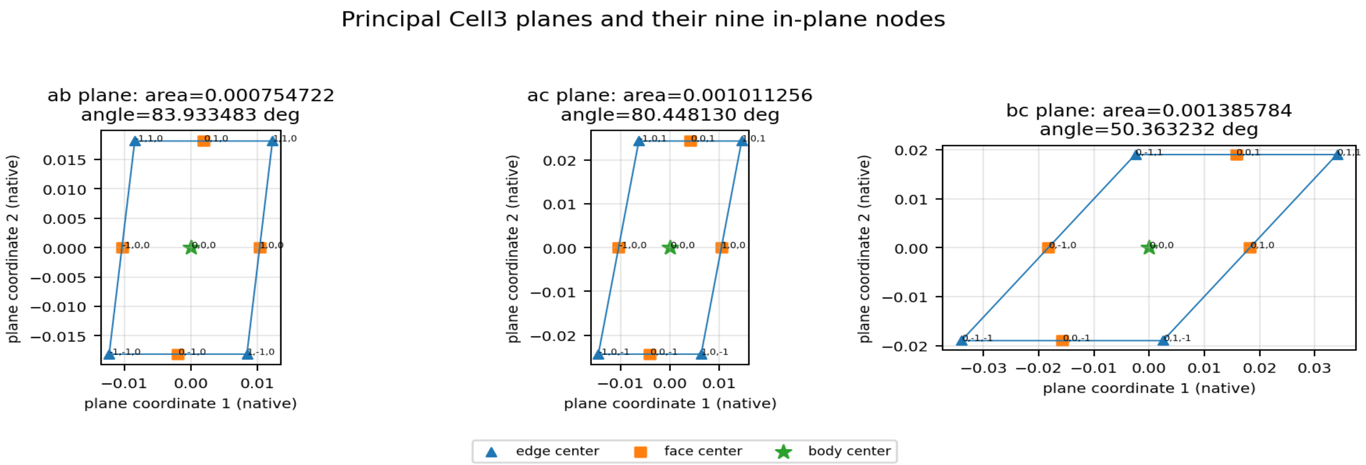

Figure 3. The ab, ac, and bc central planes. Each panel uses an orthonormal basis constructed from the corresponding source basis vectors and shows the nine nodes that lie in that plane. Face areas and included angles are source values.

Contents

1. Purpose, scope, and data boundary

This report develops the complete K-Field bullet-trajectory calculation from the archived spin-reversal experiment source and the supplied 5.56 mm ballistic table. It is internally self-sufficient: the source geometry, field transformation, projectile spin conversion, maximum lateral force, acceleration, time integration, trajectory matrix, spin-reversal separation, dimensional checks, and all Cell3 nodes are reproduced or derived explicitly.

The empirical inputs are the projectile specification and the tabulated forward velocity, energy, vertical drop, and time of flight. The K-Field source inputs are the archived Cell3 geometry, G_phase matrix, Earth velocity vector, coupling coefficient, transformed field-response vector, and locked optimum orientation. The trajectory outputs are model-derived quantities. The requested integration fixes the lateral acceleration at 0.52318 m/s^2 throughout the flight and therefore represents the optimal constant-acceleration envelope.

The 5.56 mm nominal diameter identifies the cartridge class but does not enter the reduced force law used here. In this reduction the maximum force depends on active mass and angular rate, while the direction depends on the cross product between the spin axis and the transformed field-response vector.

1.1 Source provenance

Primary source file: `K-Field_apparent_weight_modulation_spin_reversal_experiment_1784913657.json`
SHA-256: `5548ab14e1014fbd4588f407f2b8a9dd618acb1beac1f26577df9752f69ebd34`
Source schema: `K-Field.apparent_weight_modulation.spin_reversal_experiment.v1`
Source generation time: 2026-07-24T17:20:57Z

The source itself records its upstream tubular-rotor file and hash, preserving the derivation chain from free-space tubular rotation to the spin-reversal experiment.

1.2 Coordinate and sign conventions

All inertial vectors are expressed in ICRS/J2000 Cartesian coordinates. The projectile spin axis is denoted n. The transformed response is g = G_phase V_E. The lateral force follows n x g. Positive lateral displacement is defined along the source force-acceleration unit vector. Reversing spin changes n to -n and therefore reverses the lateral force and displacement without changing their magnitudes.

2. Ballistic input table

The following matrix is the supplied longitudinal ballistic solution. The zero-range marker is retained at 100 yards in ASCII-safe form. Drop is measured relative to the sight line; time of flight is the independent variable used for lateral integration.

Range	Velocity (ft/s)	Energy (ft-lb)	Drop (in)	TOF (s)
Muzzle	2910	1410	--	0.000
25 yds	2843	1346	-0.7	0.026
50 yds	2777	1285	-0.2	0.053
75 yds	2713	1226	+0.1	0.080
100 yds (zero)	2649	1169	-0.0	0.108
125 yds	2587	1115	-0.4	0.137
150 yds	2525	1062	-1.1	0.166
175 yds	2464	1011	-2.1	0.196
200 yds	2404	963	-3.5	0.227
225 yds	2345	916	-5.2	0.259
250 yds	2286	871	-7.4	0.291
275 yds	2229	827	-10.0	0.324
300 yds	2172	785	-13.0	0.359
325 yds	2116	745	-16.5	0.394
350 yds	2060	707	-20.4	0.429

Range	Velocity (ft/s)	Energy (ft-lb)	Drop (in)	TOF (s)
375 yds	2006	670	-24.9	0.466
400 yds	1953	635	-29.9	0.504
425 yds	1900	601	-35.5	0.543
450 yds	1848	569	-41.6	0.583
475 yds	1798	538	-48.5	0.624
500 yds	1748	509	-55.9	0.667

Table 1. Supplied longitudinal ballistic table used without interpolation or alteration.

3. Cell3 geometric foundation

The archived source represents the primitive cell with three non-orthogonal basis vectors a, b, and c. The basis matrix is stored with vectors as columns. The twenty-seven listed nodes occupy all integer half-step addresses (i,j,k) with each index in {-1,0,1}. Their physical coordinates are

$$r(i,j,k) = 0.5\ [a\ b\ c]\ [i\ j\ k]^T.$$

The eight addresses having no zero index are corners, twelve addresses having one zero index are edge centers, six addresses having two zero indices are face centers, and (0,0,0) is the body center.

3.1 Basis vectors and invariant dimensions

Quantity	x	y	z	Magnitude / value
a	0.020381438517	-0.003416414496	-0.002332965917	0.020797058782
b	0.002198745968	0.012902170259	-0.034066501536	0.036494205131
c	-0.000630379485	-0.024257722017	-0.042924354571	0.049308565900
primitive volume	--	--	--	2.842066937809649e-05

Table 2. Cell3 basis matrix and primitive volume in native source units.

3.2 Principal face planes

Plane	Area (native^2)	Included angle (deg)	Unit normal x	Unit normal y	Unit normal z
ab	0.000754721789673	83.933483012313	0.194092200808	0.913177168821	0.358379234791
ac	0.001011255757688	80.448130018338	0.089052594102	0.866576769291	-0.491033948326
bc	0.001385784431733	50.363231813756	-0.995965190309	0.083602162507	-0.032619290564

Table 3. Principal face areas, included angles, and ICRS unit normals.

The three areas are unequal and the included angles are non-right, establishing the anisotropic geometry that transforms the Earth velocity before the spin-force cross product is formed.

4. Field response and optimum orientation

The source uses the linear field transformation

$$g = G_{phase} V_E$$

Row	G1	G2	G3
1	2.9764106378	1.2431507795	-2.0803600354
2	1.2431507795	2.5922286635	-2.3178839937
3	-2.0803600354	-2.3178839937	6.7735709133

Table 4. Archived G_phase matrix.

Vector	x	y	z	Magnitude
V_E (m/s)	-359003.199066	76679.886034	-44701.060101	369812.502000
g = G_phase V_E	-880221.981635	-143911.436562	266335.026949	930825.217270

Table 5. Earth velocity and transformed response vectors.

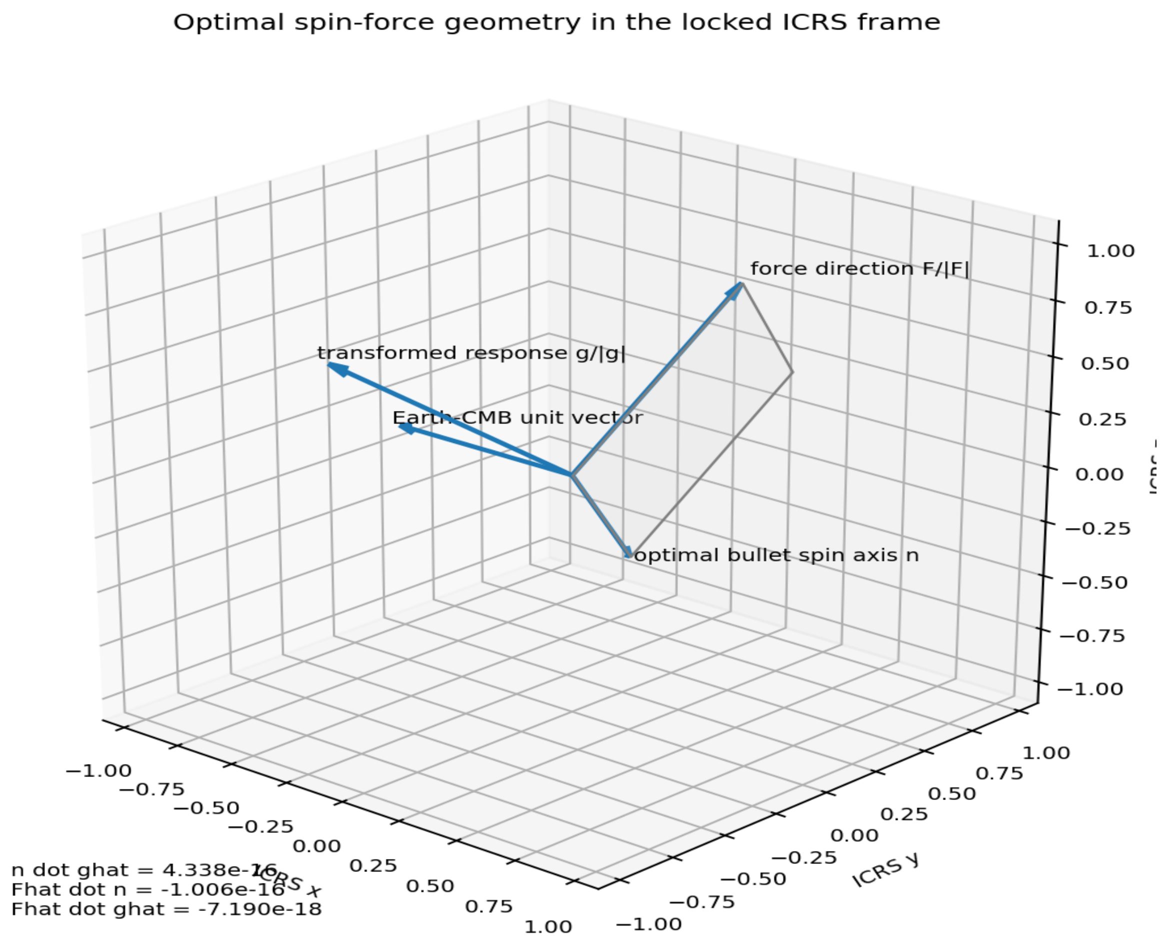

Figure 4. Locked ICRS vector geometry. Maximum force occurs when n dot g = 0; the force direction is parallel to n x g and is perpendicular to both n and g.

The archived optimum axis is n = (-0.3158650364, 0.6461626830, -0.6947683541), corresponding to RA 116.050867 deg and Dec -44.008761 deg. The resulting force direction is Fhat = (0.0774695854, 0.7473758634, 0.6598695191), corresponding to RA 84.082114 deg and Dec 41.289922 deg. The source orthogonality residual n dot g is 4.12 x 10^{-10} in its stored check, while the normalized recomputation is numerically zero to floating-point precision.

4.1 Maximum-force reduction

```
F_K = (kappa_F m omega / 2) (n x g)
```

At optimum orientation, |n x g| = |g|. Defining q_g = kappa_F |g| / 2 gives the scalar maximum-force law

```
F_max = q_g m omega.
```

The archived values are kappa_F = 3.0740508997020e-11, |g| = 930825.2172704348 m/s, and q_g = 1.4307020483078e-05 N/[kg (rad/s)].

5. Projectile spin derivation

The projectile mass is obtained directly from the grain definition:

```
m = 90 grain x 0.00006479891 kg/grain = 0.0058319019 kg.
```

The 1:6 twist means one full revolution for each 6 inches of forward travel. Therefore the spin frequency follows from the longitudinal velocity divided by the twist length:

```
f_spin = (2910 ft/s x 0.3048 m/ft) / (6 in/rev x 0.0254 m/in) = 5820.000000 rev/s.
```

```
RPM = 60 f_spin = 349200.000 rpm.
```

```
omega = 2 pi f_spin = 36568.138487785 rad/s.
```

Property	Value
Projectile mass	0.0058319019 kg = 5.8319019 g
Muzzle velocity	2910.0 ft/s = 886.968 m/s
Twist	6.0 in/rev = 0.1524 m/rev
Spin frequency	5820.000000 rev/s
Spin rate	349200.000 rpm
Angular rate	36568.138487785 rad/s

Table 6. Projectile spin state at the muzzle.

6. Maximum lateral force and acceleration

Substituting the projectile mass and angular rate into the optimum scalar law gives

```
F_max = (1.4307020483078e-05)(0.0058319019)(36568.138487785) = 0.003051140888 N.
```

Maximum one-spin lateral force: 3.051141 mN.

The coupling interval stored in the source propagates to 3.020063 to 3.082218 mN.

The force divided by projectile mass gives

a_perp = F_max / m = 0.523181106 m/s^2.

The trajectory integration uses the requested rounded value a_perp = 0.52318 m/s^2 = 1.71646982 ft/s^2 = 0.05334951 g.

The projectile weight is 57.191421 mN, so the maximum lateral K-Field force is 5.334963% of projectile weight. Reversing spin changes the sign of the 3.051141 mN force and creates a full force separation of 6.102282 mN between opposite spin senses.

7. Time-of-flight integration

With zero initial lateral velocity and the specified constant lateral acceleration, elementary translational kinematics gives

v_perp(t) = a_perp t

d_perp(t) = 0.5 a_perp t^2.

These equations are evaluated independently at every supplied time-of-flight sample. Forward speed and energy do not enter this fixed-acceleration integration; they remain in the matrix to preserve the ballistic context. The displacement grows quadratically with time, while lateral velocity grows linearly.

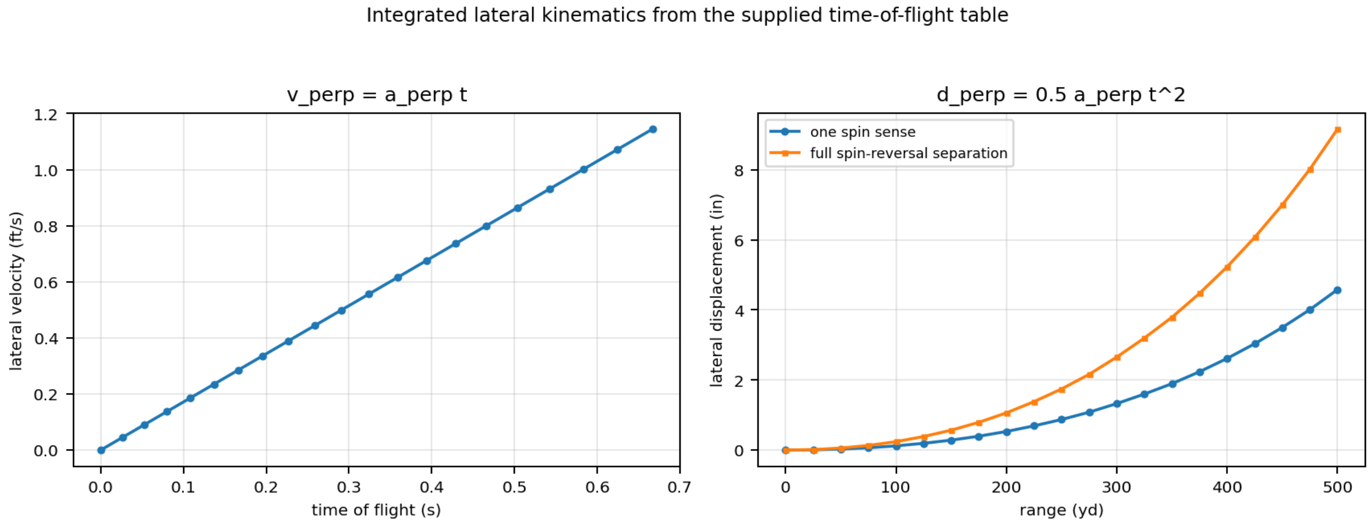

Figure 5. Lateral velocity, one-spin displacement, and full spin-reversal track separation evaluated at the supplied time samples.

8. Complete compact trajectory matrix

The matrix below reproduces the complete ballistic input and the derived lateral velocity and cumulative lateral distance for every time step. All derived values use a_perp = 1.7164698163 ft/s^2.

Range	Forward velocity (ft/s)	Energy (ft-lb)	Drop (in)	TOF (s)	Lateral velocity (ft/s)	Lateral distance (ft)	Lateral distance (in)	Full reversal separation (in)
Muzzle	2910	1410	--	0.000	0.0000	0.0000	0.000	0.000
25 yds	2843	1346	-0.7	0.026	0.0446	0.0006	0.007	0.014
50 yds	2777	1285	-0.2	0.053	0.0910	0.0024	0.029	0.058
75 yds	2713	1226	+0.1	0.080	0.1373	0.0055	0.066	0.132
100 yds (zero)	2649	1169	-0.0	0.108	0.1854	0.0100	0.120	0.240
125 yds	2587	1115	-0.4	0.137	0.2352	0.0161	0.193	0.387
150 yds	2525	1062	-1.1	0.166	0.2849	0.0236	0.284	0.568
175 yds	2464	1011	-2.1	0.196	0.3364	0.0330	0.396	0.791
200 yds	2404	963	-3.5	0.227	0.3896	0.0442	0.531	1.061
225 yds	2345	916	-5.2	0.259	0.4446	0.0576	0.691	1.382
250 yds	2286	871	-7.4	0.291	0.4995	0.0727	0.872	1.744
275 yds	2229	827	-10.0	0.324	0.5561	0.0901	1.081	2.162
300 yds	2172	785	-13.0	0.359	0.6162	0.1106	1.327	2.655
325 yds	2116	745	-16.5	0.394	0.6763	0.1332	1.599	3.197
350 yds	2060	707	-20.4	0.429	0.7364	0.1580	1.895	3.791
375 yds	2006	670	-24.9	0.466	0.7999	0.1864	2.236	4.473
400 yds	1953	635	-29.9	0.504	0.8651	0.2180	2.616	5.232
425 yds	1900	601	-35.5	0.543	0.9320	0.2530	3.037	6.073
450 yds	1848	569	-41.6	0.583	1.0007	0.2917	3.500	7.001
475 yds	1798	538	-48.5	0.624	1.0711	0.3342	4.010	8.020
500 yds	1748	509	-55.9	0.667	1.1449	0.3818	4.582	9.164

Table 7. Complete ballistic and lateral-trajectory matrix. The one-spin lateral distance is measured from the unforced longitudinal trajectory; opposite spin senses lie on opposite sides, so their separation is twice the one-spin displacement.

9. Direct numerical result

Using a_perp = 0.52318 m/s^2 = 1.71647 ft/s^2, the endpoint at 500 yards and 0.667 s is

```
v_perp = a_perp t = 1.1449 ft/s
```

$$d\_perp = 0.5\ a\_perp\ t^2 = 0.3818\ ft = 4.582\ in.$$

At 500 yards: lateral velocity = 1.1449 ft/s; one-spin lateral displacement = 4.582 in; opposite-spin track separation = 9.164 in.

The direct lateral-velocity angle relative to the local forward velocity at 500 yards is 0.037527 deg. This angle describes the instantaneous velocity vector, not the accumulated position offset.

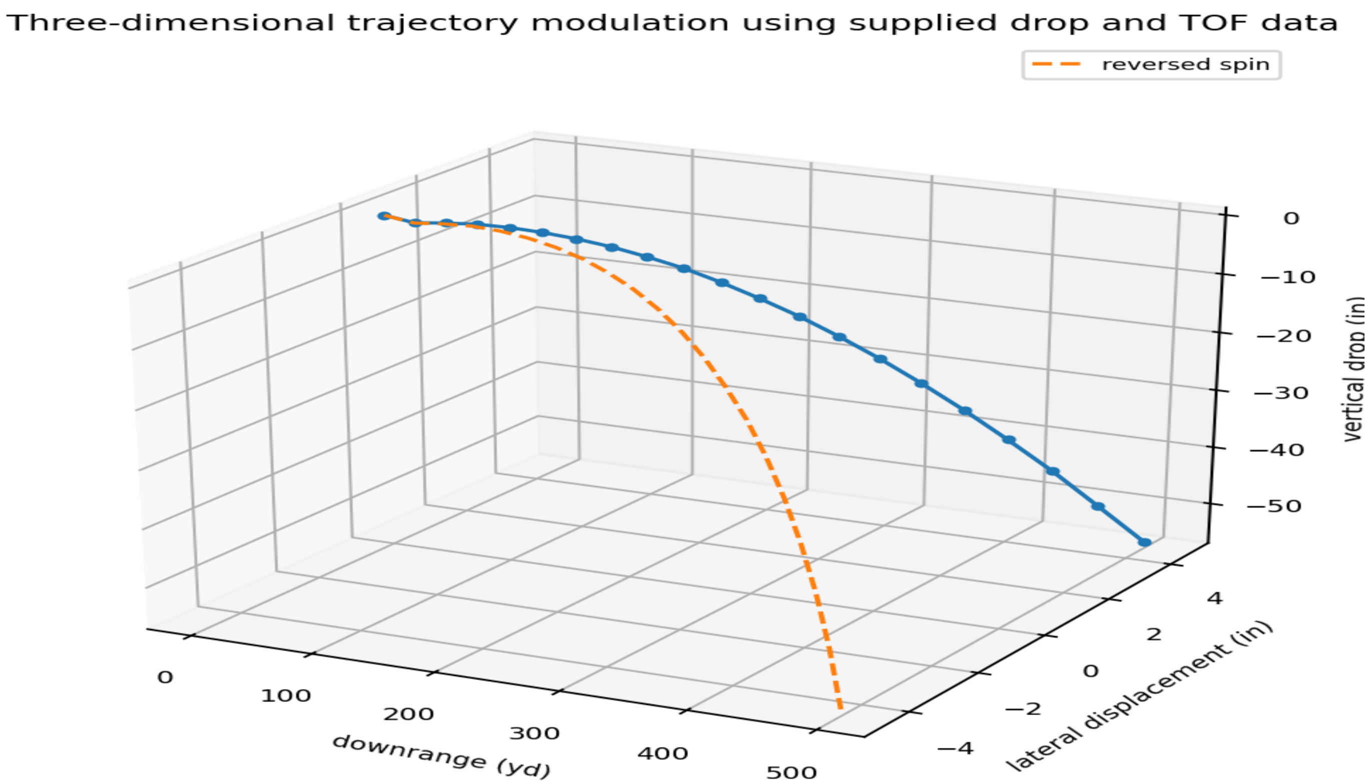

Figure 6. Three-dimensional representation combining the supplied downrange distance and vertical drop with the derived lateral displacement. The dashed trajectory is the reversed-spin image.

10. Range evolution and spin-reversal symmetry

The lateral displacement is small at short range because time appears quadratically. At 100 yards, the time of flight is 0.108 s and the one-spin displacement is 0.120 in. At 250 yards, 0.291 s produces 0.872 in. At 400 yards, 0.504 s produces 2.616 in. At 500 yards, 0.667 s produces 4.582 in.

Spin reversal is exactly antisymmetric in the source law. If the original spin gives +d_perp, reversed spin gives -d_perp. Consequently, a paired comparison between equal-magnitude opposite-spin projectiles doubles the observable spatial separation while cancelling any spin-even common trajectory component in the idealized model.

10.1 Relation to the supplied vertical drop

At 500 yards the supplied vertical drop magnitude is 55.9 in and the predicted one-spin lateral displacement is 4.582 in. The lateral magnitude is therefore 8.196% of the tabulated vertical-drop magnitude. The two quantities act along orthogonal axes and should not be algebraically combined without specifying a sight and coordinate convention.

10.2 Orientation envelope

The reported values are maxima. For a general angle beta between the bullet spin axis and g, the force magnitude is reduced by sin(beta):

```
F(beta) = F_max sin(beta).
```

A laboratory or field trajectory therefore acquires the full result only when the bullet axis is perpendicular to the transformed response vector. Components along a selected local lateral direction require an additional directional projection of n x g onto that axis.

11. Dimensional analysis and numerical validation

The source coefficient q_g has units N/[kg (rad/s)]. Radians are dimensionless, so q_g m omega has units N. Dividing by mass gives m/s^2. Multiplying acceleration by time gives velocity, and multiplying acceleration by time squared gives distance. The conversion 1 m = 3.280839895 ft and 1 ft = 12 in is applied after the SI acceleration is fixed.

The following identities were recomputed directly from source arrays and report inputs:

Check	Computed value	Acceptance
All Cell3 nodes present	27	27 required
Basis a length	0.020797058782	0.020797058782
Basis b length	0.036494205131	0.036494205131
Basis c length	0.049308565900	0.049308565900
|G_phase V_E|	930825.217270435	930825.217270435
n dot ghat	4.338e-16	approximately 0
Fhat dot n	-1.006e-16	approximately 0
Fhat dot ghat	-7.190e-18	approximately 0
500 yd v_perp	1.144885367 ft/s	a t
500 yd d_perp	4.581831241 in	0.5 a t^2

Table 8. Independent dimensional and numerical validation checks.

The 0.52318 m/s^2 value differs from the unrounded force-derived acceleration by approximately 1.1×10^{-6} m/s^2. The report follows the explicitly requested rounded acceleration for every trajectory entry, ensuring exact reproducibility of the compact matrix.

Appendix A. Complete Cell3 node registry

All twenty-seven half-step nodes from the source file are listed below. Coordinates are ICRS native values and integer addresses follow the source order.

i	j	k	x native	y native	z native	node class	i+j+k
-1	-1	-1	-0.010974902499961	0.007385983127423	0.039661911011939	corner	-3
-1	-1	0	-0.011290092242635	-0.004742877881141	0.018199733726392	edge center	-2
-1	-1	1	-0.011605281985309	-0.016871738889705	-0.003262443559155	corner	-1
-1	0	-1	-0.009875529515800	0.013837068256769	0.022628660243911	edge center	-2
-1	0	0	-0.010190719258473	0.001708207248205	0.001166482958364	face center	-1
-1	0	1	-0.010505909001147	-0.010420653760359	-0.020295694327182	edge center	0
-1	1	-1	-0.008776156531638	0.020288153386115	0.005595409475884	corner	-1
-1	1	0	-0.009091346274312	0.008159292377552	-0.015866767809663	edge center	0
-1	1	1	-0.009406536016985	-0.003969568631012	-0.037328945095210	corner	1
0	-1	-1	-0.000784183241488	0.005677775879218	0.038495428053575	edge center	-2
0	-1	0	-0.001099372984162	-0.006451085129346	0.017033250768028	face center	-1
0	-1	1	-0.001414562726835	-0.018579946137910	-0.004428926517519	edge center	0
0	0	-1	0.000315189742674	0.012128861008564	0.021462177285547	face center	-1
0	0	0	0.000000000000000	0.000000000000000	0.000000000000000	body center	0
0	0	1	-0.000315189742674	-0.012128861008564	-0.021462177285547	face center	1
0	1	-1	0.001414562726835	0.018579946137910	0.004428926517519	edge center	0
0	1	0	0.001099372984162	0.006451085129346	-0.017033250768028	face center	1
0	1	1	0.000784183241488	-0.005677775879218	-0.038495428053575	edge center	2
1	-1	-1	0.009406536016985	0.003969568631012	0.037328945095210	corner	-1
1	-1	0	0.009091346274312	-0.008159292377552	0.015866767809663	edge center	0
1	-1	1	0.008776156531638	-0.020288153386115	-0.005595409475884	corner	1
1	0	-1	0.010505909001147	0.010420653760359	0.020295694327182	edge center	0
1	0	0	0.010190719258473	-0.001708207248205	-0.001166482958364	face center	1
1	0	1	0.009875529515800	-0.013837068256769	-0.022628660243911	edge center	2
1	1	-1	0.011605281985309	0.016871738889705	0.003262443559155	corner	1
1	1	0	0.011290092242635	0.004742877881141	-0.018199733726392	edge center	2
1	1	1	0.010974902499961	-0.007385983127423	-0.039661911011939	corner	3

Table A1. Complete authoritative Cell3 node registry.

Appendix B. Reproducible equation set

The complete scalar calculation can be reproduced from the following sequence:

1. `m = 90 x 0.00006479891 kg`
2. `v = 2910 x 0.3048 m/s`
3. `p_twist = 6 x 0.0254 m/rev`
4. `f_spin = v / p_twist`
5. `omega = 2 pi f_spin`
6. `g = G_phase V_E`
7. `q_g = kappa_F |g| / 2`
8. `F_max = q_g m omega`
9. `a_perp = F_max / m`
10. `For the requested trajectory table, set a_perp = 0.52318 m/s^2`
11. `v_perp(t_i) = a_perp t_i`
12. `d_perp(t_i) = 0.5 a_perp t_i^2`
13. `opposite-spin separation(t_i) = 2 d_perp(t_i).`

Appendix C. Machine-readable companion

The companion file `K-Field_bullet_trajectory_modulation_1785151191.json` contains the complete source identifiers, geometry snapshot, field parameters, projectile inputs, every trajectory row, endpoint results, equations, and validation values. Numeric values are stored as JSON numbers in SI and customary units where used in the input table.

End of archival reference.